

Linux Process Management – Hands-on Tasks

Task 1: View Running Processes

Objective: Learn how to view running processes.

Steps

1. Open the terminal.
2. Run the following commands:

```
ps  
ps -ef  
ps aux
```

Task

- Identify:
 - PID
 - User
 - Command

Question

- Which process has **PID 1**?

Task 2: Monitor Processes in Real Time

Objective: Monitor system processes.

Steps

Run:

top

Tasks

- Identify:
 - CPU usage
 - Memory usage
 - Load average

Bonus

Press:

q

to exit.

Task 3: Install and Use htop

Objective: Use an advanced process viewer.

Steps

Install:

sudo yum install htop

or

sudo apt install htop

Run:

htop

Tasks

- Sort processes by **CPU usage**
 - Kill a process using the interface
-

Task 4: Run a Process in Background

Objective: Understand background processes.

Steps

Run:

```
sleep 100 &
```

Check jobs:

```
jobs
```

Tasks

- Note the **job ID**
 - Note the **PID**
-

Task 5: Bring Background Job to Foreground

Objective: Practice job control.

Run:

```
sleep 200 &
```

Check:

```
jobs
```

Bring to foreground:

`fg %1`

Task 6: Stop and Resume Processes

Run:

`sleep 300`

Press:

`CTRL + Z`

Resume in background:

`bg`

Check jobs:

`jobs`

Task 7: Find Process ID

Objective: Identify process PID.

Run:

`sleep 500 &`

Find PID:

`pidof sleep`

or

`pgrep sleep`

Task 8: View Process Tree

Objective: Understand parent-child processes.

Run:

ps tree

Task

Find the hierarchy for:

systemd

Example observation:

systemd → sshd → bash

Task 9: Kill a Process

Objective: Terminate processes.

Run:

sleep 500 &

Find PID:

ps aux | grep sleep

Kill process:

kill PID

Verify:

ps aux | grep sleep

Task 10: Force Kill a Process

Run:

sleep 500 &

Kill using SIGKILL:

kill -9 PID

Task 11: Kill Process by Name

Run multiple processes:

```
sleep 500 &  
sleep 500 &  
sleep 500 &
```

Kill all:

```
killall sleep
```

Task 12: Use pkill

Run:

```
sleep 600 &
```

Kill using pattern:

```
pkill sleep
```

Task 13: Run Process with Priority

Objective: Use nice values.

Run:

```
nice -n 10 sleep 200
```

Check priority:

```
top
```

Task 14: Change Priority of Running Process

Run:

sleep 400 &

Find PID:

ps aux | grep sleep

Change priority:

renice -n 5 -p PID

Task 15: Monitor System Load

Run:

uptime

Task

Observe:

Load average

Task 16: Run Command Repeatedly

Run:

watch -n 2 ps

Task

Observe process changes every **2 seconds**.

Exit:

CTRL + C

Task 17: Detect Zombie Processes

Run:

```
ps aux | grep Z
```

Task

Identify if any process shows:

Z

Task 18: Check Web Server Processes

Run:

```
ps aux | grep nginx
```

or

```
ps aux | grep apache
```

Task

Identify:

- Master process
 - Worker processes
-

Task 19: Background Backup Task

Run:

```
tar -czf backup.tar.gz /etc &
```

Check:

```
jobs
```


Monitor:

```
ps aux | grep tar
```

Task 20: Investigate High CPU Process

Run:

```
top
```

Identify a high CPU process.

Investigate:

```
ps -fp PID
```

Kill if needed:

```
kill PID
```
